

CT214H – Web Programming Fundamentals

Chapter 1

Internet, WWW and HTML

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Content

- Internet
- World Wide Web (WWW)
- HyperText Markup Language (HTML)

The Internet

What the Internet is?

- A **world-wide network** of computer networks (a network of networks)
- At the lowest level, since 1982, all connections use **TCP/IP**
- TCP/IP hides the differences among devices connected to the Internet
- **Origin** of the Internet:
 - ARPAnet - late 1960s and early 1970s (for military uses)
 - Network reliability
 - For ARPA-funded research organizations

The Internet basic concepts

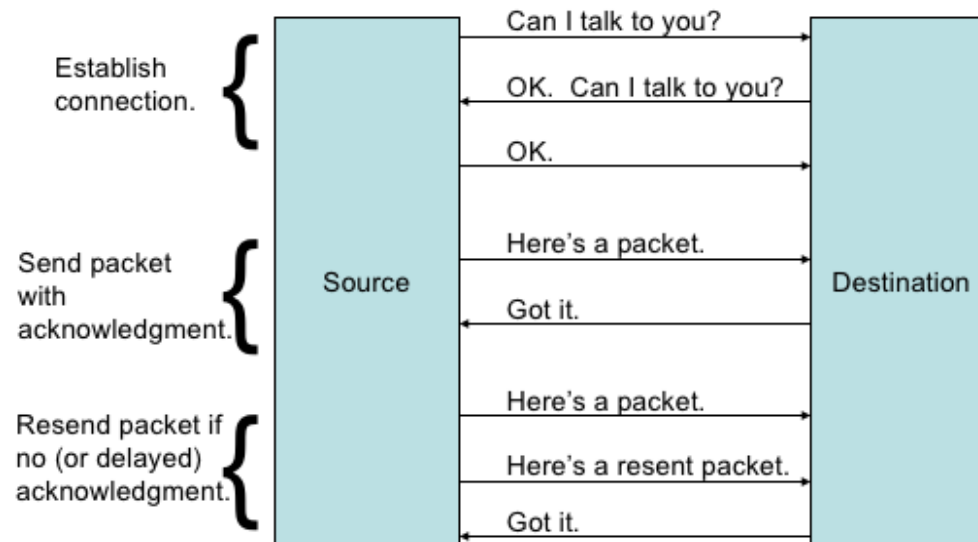
- The Internet
- Communication protocols
- IP (Internet protocol)
 - What is IP?
 - IPv4 and IP v6
- TCP (Transmission Control Protocol)
- UDP (User Datagram Protocol)
- DNS (Domain Name System)

The Internet basic concepts

- **Communication protocol**: how computers talk?
 - telephone “protocol”: how you answer and end a call, what language you speak, etc
- **Internet Protocol (IP)**:
 - A protocol to **address computers** on Internet (every node has a unique numeric address)
 - **32-bit** number (in IPv4) or **128-bit** number (in IPv6)
 - Associated with at most one device at a time (although the device may have more than one)
 - Written as four dot-separated bytes, e.g., 192.168.1.123 (IPv4)

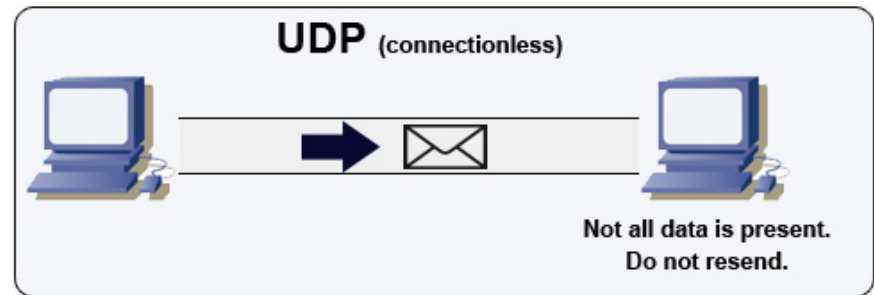
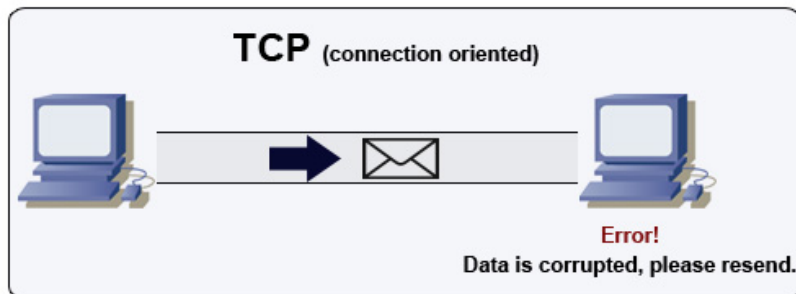
The Internet basic concepts

- **TCP** (Transmission Control Protocol): adds the concept of a **connection** and **port** on top of the IP
 - Provides **guarantee** that packets delivered
 - Provide two-way (full duplex) communication



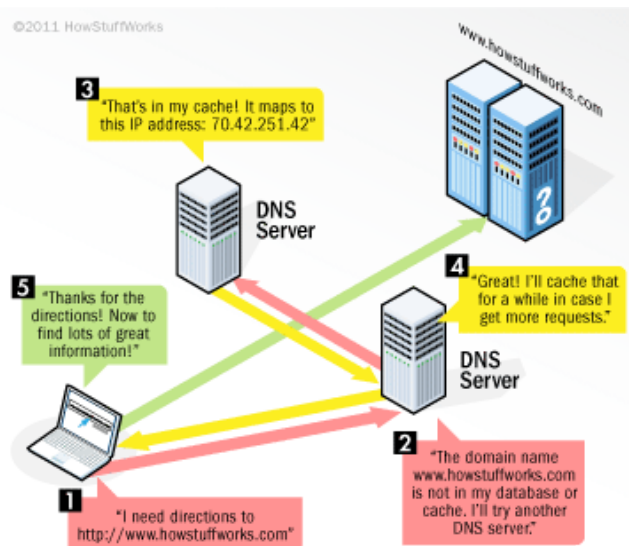
The Internet basic concepts

- **UDP** (User Datagram Protocol):
 - Builds on IP and provides port concept
 - No connection concept
 - No transmission guarantee
 - **Lightweight**, so faster for one-time messages



The Internet basic concepts

- **DNS** (Domain Name System):
 - Form: host.domain, for example: www.ctu.edu.vn
 - **First** domain is the **smallest**, **last** is the **largest** (top level domain)
 - Last domain specifies the **type** (or **location**) of the **organization**
 - **DNS servers**: convert DNS to IP



Analogy to Telephone Network

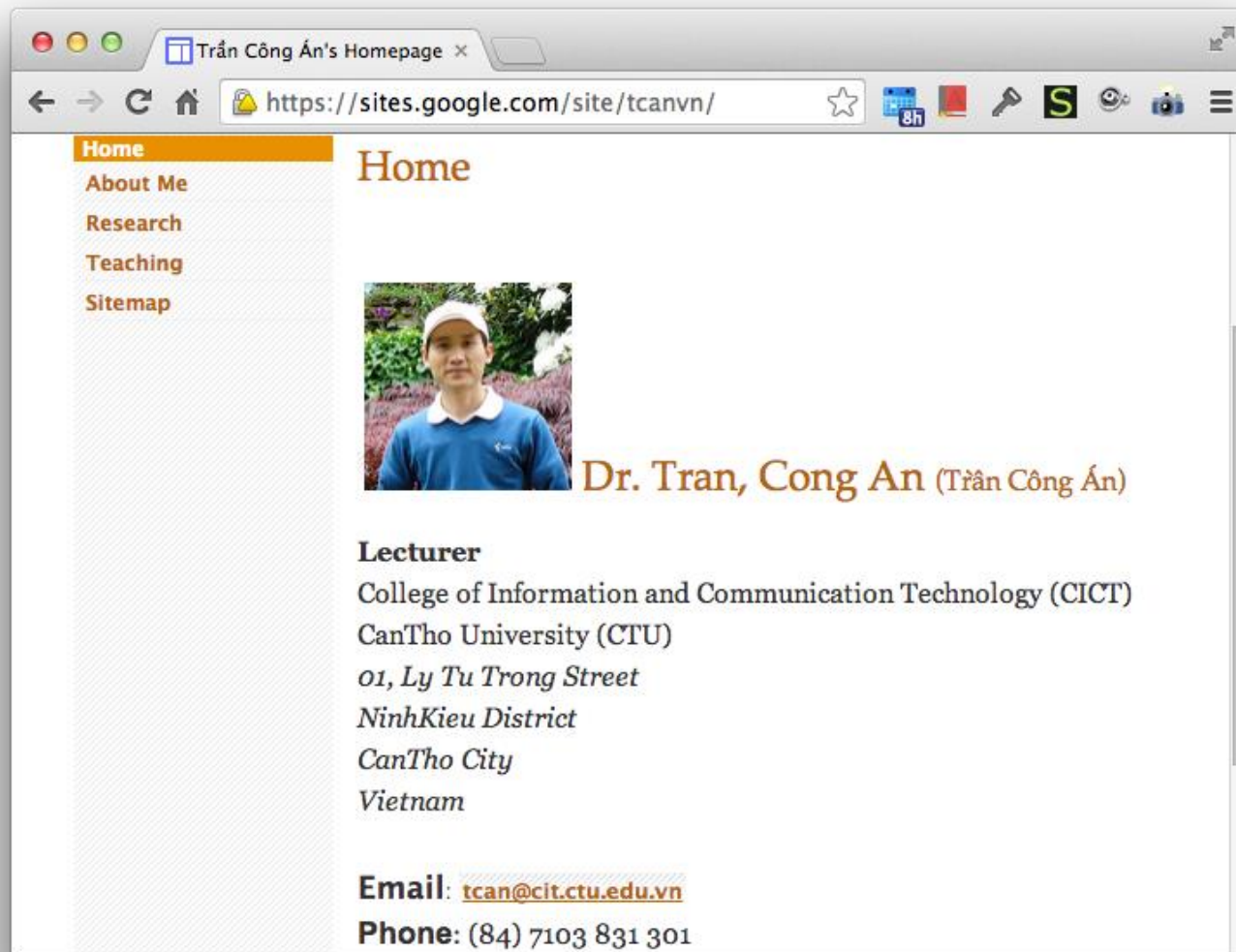
- IP ~ the telephone network
- TCP ~ calling someone who answers, having a conversation, and hanging up
- UDP ~ calling someone and leaving a message
- DNS ~ directory assistance

The WWW

What is WWW?

- One of services for organizing Internet-based information
 - Document form: HyperText
 - Information representation using HTML (HyperText Markup Language)
 - Communication protocol: HTTP (HyperText Transfer Protocol)
- Origins:
 - Tim Berners-Lee at CERN (European Organization for Nuclear Research) in 1989
 - Purpose: to allow scientists to have access to many databases of scientific work (share) through their own computers

WWW Document

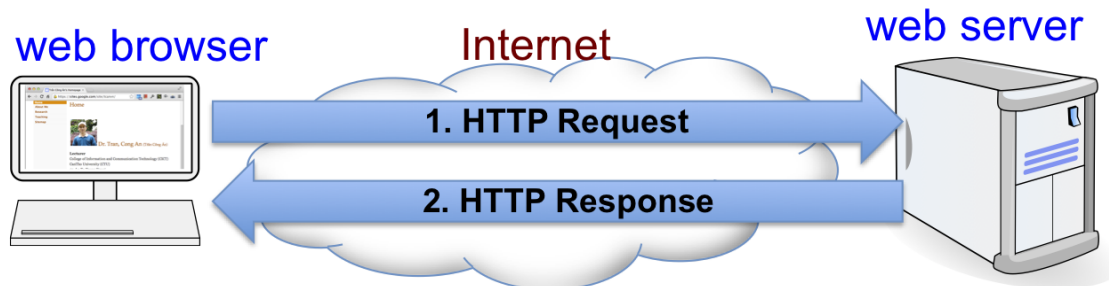


Hypertext, Webpage and Website

- **HyperText**: text displayed on a computer display, which contains links to other texts (hyperlinks)
- **Webpage**:
 - a document commonly written in HTML
 - accessible through the Internet or other networks using an Internet browser
- **Website**:
 - a collection of related web pages
 - typically identified with a common domain name, and published on at least one web server.

Web Server and Web Browser

- **The Web** is a collection of **Web Servers** on the Internet that provides HTTP documents via HTTP protocol, either existing documents (static web) or dynamically built documents (dynamic web)
 - Some popular web servers: Apache (open source), IIS (Windows)
- A **Web Browser** is a computer program that is used by end users to access the Web
 - Popular web browsers: Chrome, Firefox, Microsoft Edge,...

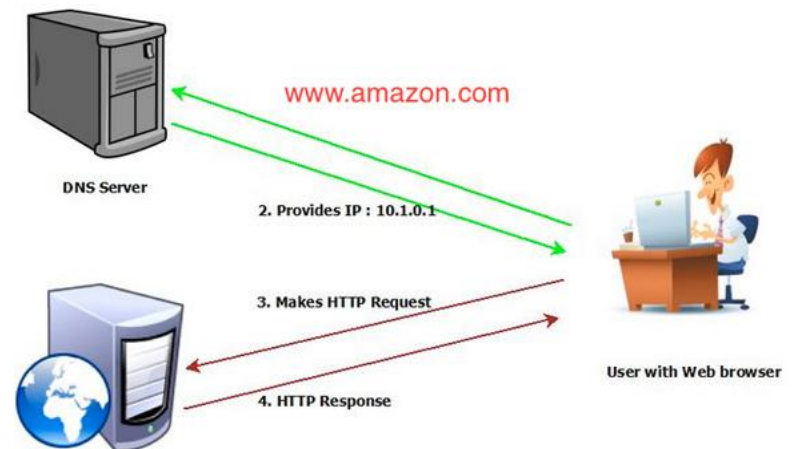


Web Server and Web Browser

- Additional **functionality** of Web Browsers:
 - Execution of **scripts** (e.g., drop-down menus)
 - **Event** handling (e.g., mouse clicks)
 - GUI for **controls** (e.g., buttons)
 - **Secure communication** with servers
 - Display of non-HTML documents (e.g., PDF) via **plug-ins**

Web Browser and Web Server

- Typical browser-server interaction:
 - 1) User enters Web **address** in browser (DNS)
 - 2) Browser uses **DNS** to locate **IP** address
 - 3) Browser opens **TCP connection** to server
 - 4) **Browser** sends **HTTP request** over connection
 - 5) **Server** sends **HTTP response** to browser over connection
 - 6) Browser **displays** body of response in the client area of the browser window

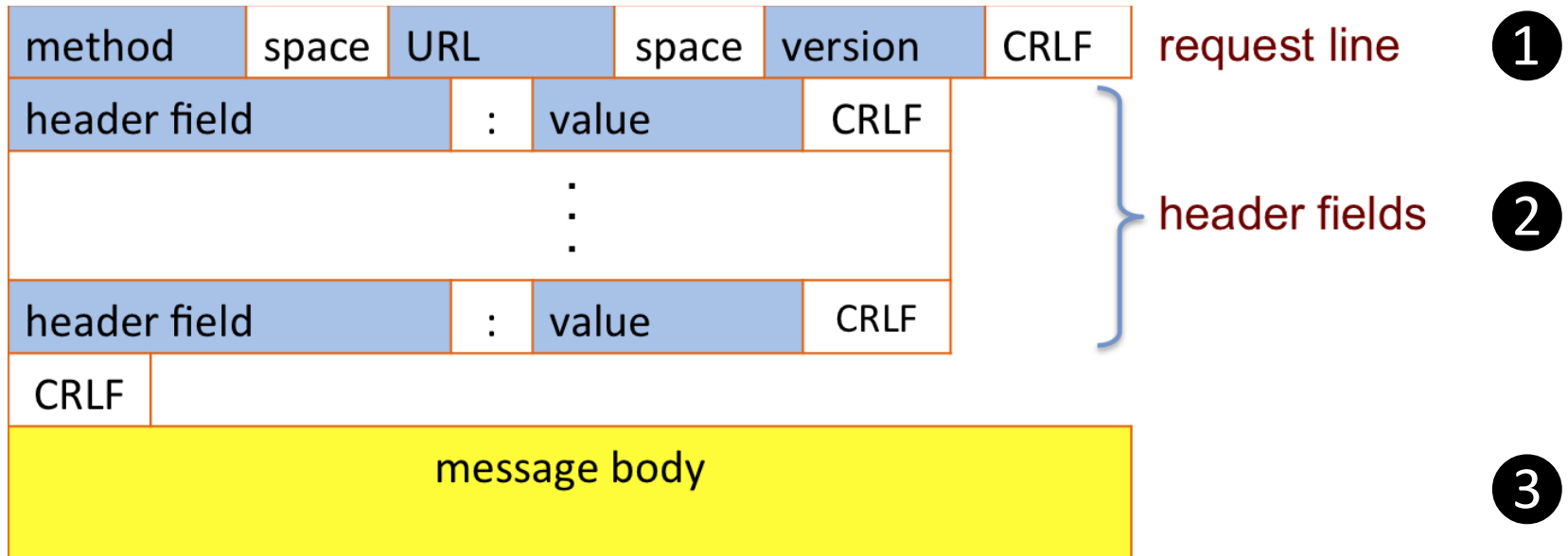


HyperText Transfer Protocol (HTTP)

- **Communication protocol** between browsers and web servers
 - A set of rules that are used in communication between web browsers and web servers
- HTTP is a **stateless** protocol:
 - The protocol does not require the server to remember anything about the client between requests
- Normally implemented over **TCP protocol**
- Information transmitted: **HyperText** (plant text)
- HTTP versions: 0.9, 1.0, 1.1 (currently)

HyperText Transfer Protocol (HTTP)

Structure of a **HTTP request** package



HyperText Transfer Protocol (HTTP)

1) Request line:

- method: GET, POST, HEAD, PUT, DELETE,...
- URL: path to the requested resource
- version: HTTP version used in communication between client and server (currently 1.1)
- Example: **GET / HTTP/1.1**
(request the homepage using HTTP protocol version 1.1)

2) Header fields: additional information with the request

3) Body: usually empty, except in case browser wants to upload

HyperText Transfer Protocol (HTTP)

- Request methods (case sensitive):
 - **GET**: request a resource from the server given a URL
 - **POST**: used to send data to server (using HTML forms)
 - **HEAD**: same as GET method but get the status and header section only
 - **PUT**: replaces resources identified by a URL with the uploaded content
 - **DELETE**: removes resources given by a URL

Note: Not all request methods are allowed by a web server

HyperText Transfer Protocol (HTTP)

- URL:

- Specifies location at which a resource can be found
- HTTP URL Syntax:

http://host_name/path/command?parameter1=value¶meter2...

- Example of HTTP URL:

http://www.example.org:56789/a/b/c.txt?t=win&s=chess#para5

The diagram shows the URL `http://www.example.org:56789/a/b/c.txt?t=win&s=chess#para5` with arrows pointing to its components: `www.example.org` is labeled "host (FQDN)", `56789` is labeled "port", `/a/b/c.txt` is labeled "path", `?t=win&s=chess` is labeled "query", and `#para5` is labeled "fragment". A blue horizontal line underlines the `/a/b/c.txt?t=win&s=chess#para5` portion, with the label "Request-URI" centered below it.

HyperText Transfer Protocol (HTTP)

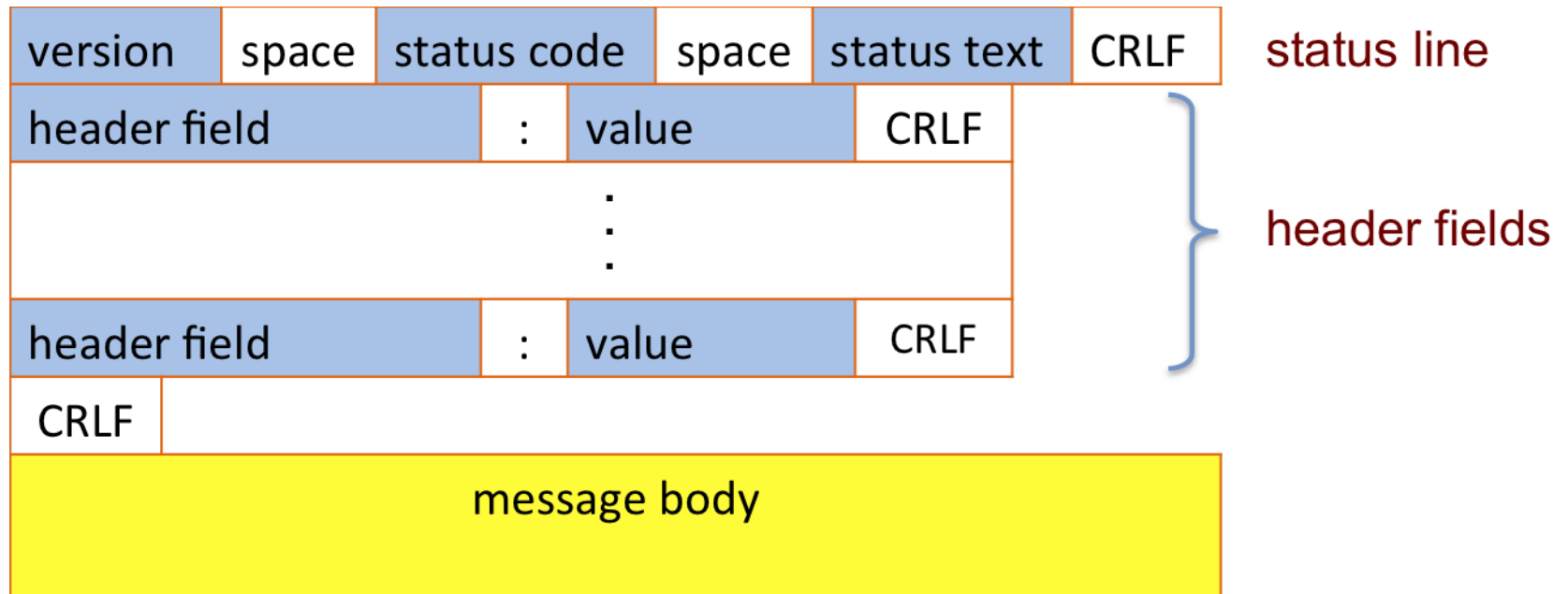
- Header fields: case insensitive

field name : field value

- Host: host name from URL (required)
 - User-Agent: type of browser sending request
 - Accept: **MIME** types of acceptable documents (e.g text/html, image/jpg)
 - Content-Length: bytes in body
 - Content-Encoding: codings have been applied to the entity-body
 - Accept-Language: languages that are understandable by browser
- MINE: Convention for specifying **content type** of a message

HyperText Transfer Protocol (HTTP)

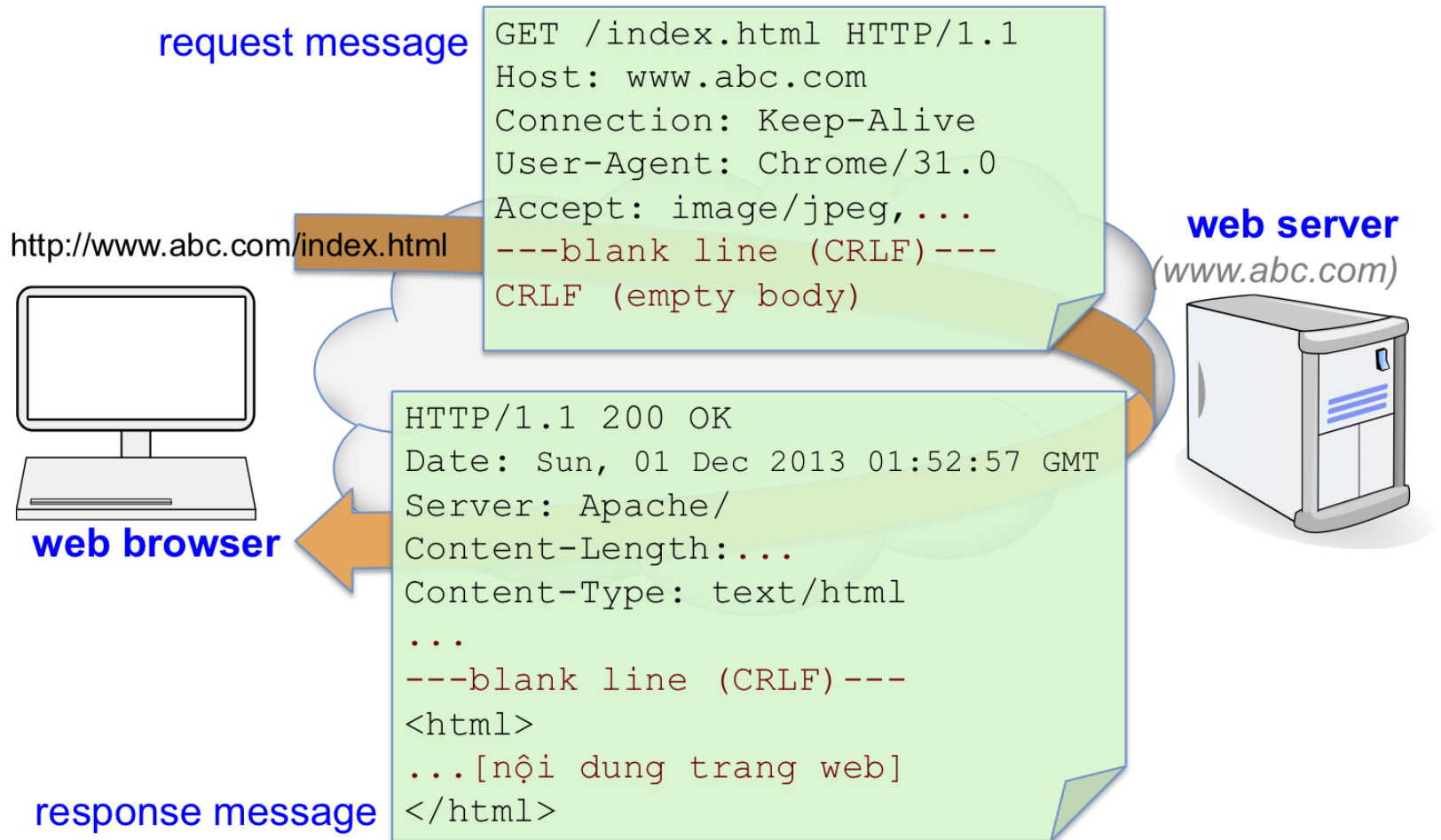
Structure of a **HTTP response** package:



HyperText Transfer Protocol (HTTP)

- Status: contains the status of the response
 - HTTP version
 - Status code: three digits
 - First digit: 1=information; 2=success; 3=redirection; 4=client error; 5=server error
 - Other two digits: additional information
 - Status text: reason phrase (for human use)
- Header fields:
 - Date: date and time at which response was generated (required)
 - Location: alternate URI if status is redirection
 - Last-Modified: date and time the requested resource was last modified

Example of Request/Response method



WWW Resources

- Web pages
- Multimedia data: images, sound, videos, etc.

The screenshot shows a web browser window displaying a TED talk page. The address bar shows the URL: www.ted.com/talks/henry_evans_and_chad_jenkins_meet_the_robots_for_humanity.html. The TED logo and tagline "Ideas worth spreading" are at the top left. A navigation menu on the top right includes links for Talks, Speakers, Playlists, Translations, TED Conferences, TEDx Events, TED Prize, TED Fellows, TED Conversations, TED Community, TED-Ed, About TED, TED Blog, and TED Initiatives. A search bar is located below the menu. A banner for "New TED Talks" with a subscription form and social media links is present. The main content area features the title "TALKS | TEDX Henry Evans and Chad Jenkins: Meet the robots for humanity" and the text "FILMED OCT 2013 • POSTED NOV 2013 • TEDxMidAtlantic". A video player shows a scene from the talk with a robot on a stage. To the right of the video, the view count is "262,980 Views" and there are "8.5K" likes. A description of the talk follows, mentioning Henry Evans' use of a telepresence robot and a quadrotor drone. A "Full bio" link is provided. At the bottom, there is a "RELATED PLAYLISTS" section with a link to "Will drones save us or destroy us?".

Henry Evans and Chad Jenkins: Meet the robots for humanity

FILMED OCT 2013 • POSTED NOV 2013 • TEDxMidAtlantic

262,980 Views 8.5K

Paralyzed by a stroke, Henry Evans uses a telepresence robot to take the stage – and show how new robotics, tweaked and personalized by a group called Robots for Humanity, help him live his life. He shows off a nimble little quadrotor drone, created by a team led by Chad Jenkins, that gives him the ability to navigate space – to once again look around a garden, stroll a campus ... (Filmed at TEDxMidAtlantic.)

In 2003, Henry Evans became quadriplegic and mute after a stroke-like attack. Now, working with Robots for Humanity, he's a pioneer in adaptive robotic tech to help him, and other disabled people like him, navigate the world. [Full bio »](#)

RELATED PLAYLISTS **NEW** [View more »](#)

Will drones save us or destroy us?

Uniform Resource Locator (URL)

- Used to **identify a resource** on the web

- Syntax:

`<protocol>://<host address>[:port]/<path to the resource>`

- Examples:

- <http://www.w3.org/Consortium/siteindex.html>
- <ftp://ctan.tug.org/pub/tex/tds.zip>
- <mailto:tcan@cit.ctu.edu.vn>
- <file:///c:/Windows/clock.avi>

Uniform Resource Locator (URL)

- Absolute URL:

- Fully qualified URL which contains **all elements** of an URL
- Example: <http://www.w3.org/Consortium/siteindex.html>

- Relative URL:

- Contains only the **relative path** to the resource
- Protocol and host address are inferred using the information in the URL of the **containing page**
- **Absolute path** of the resource is calculated from the address (path) of **containing the webpage**
- Example: images/logo.png

Types of web

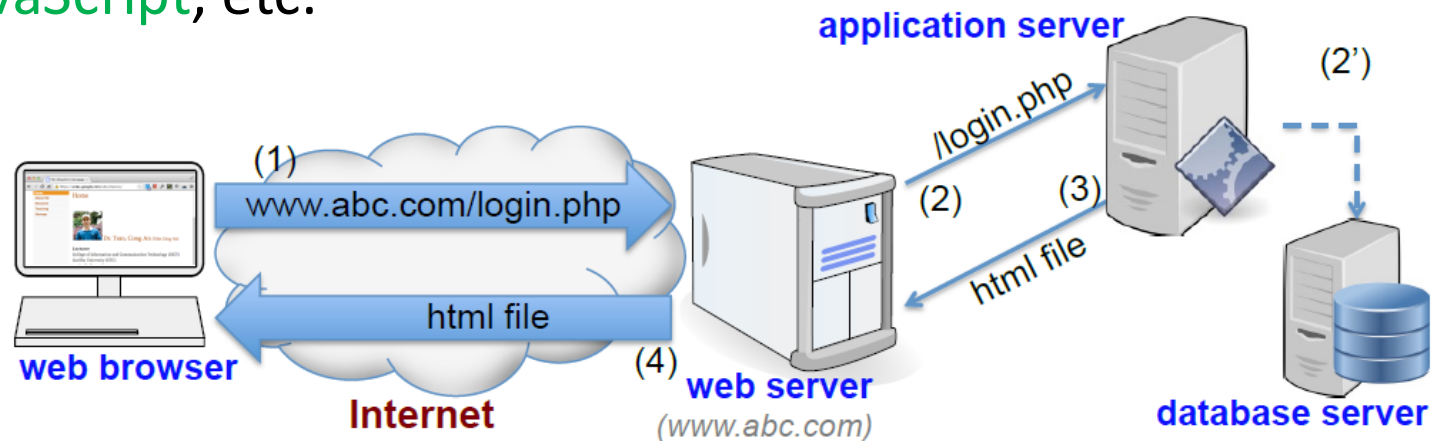
- **Static web:**
 - Content of the webpage is **fixed**
 - Extension: **.html**
 - Can be served directly by web server
- **Dynamic web:**
 - Content of a webpage is “**dynamic**”
 - Usually **produced** by a **programming/scripting language**

Client-side scripting languages

- Embedded in the webpage
- Executed by browser
- Can be used to:
 - Interact with the web users
 - Validate data at the browser
 - Modify the web content or interface accordingly the context
 - Communicate with server
- Some client-side scripting languages: JavaScript, ActionScript, VBScript, etc.

Server-side scripting languages

- Used on the **server side**
- Do the business logic
- The result is usually a webpage
- Execute by an application server
- Some server-side scripting languages: ASP(.NET), JSP, PHP, **JavaScript**, etc.



HTML

(HyperText Markup Language)

Introduction to HTML

- HTML is the standard markup language for creating Web pages
- HTML describes the content and structure of Web pages using markup
- This is not a programming language!

- ▶ 1991: HTML
- ▶ 1994: HTML 2
- ▶ 1996: CSS1 + JavaScript
- ▶ 1997: HTML4 (*)
- ▶ 1998: CSS2
- ▶ 2000: XHTML 1
- ▶ 2002: Tableless Web Design
- ▶ 2005: AJAX
- ▶ 2009: HTML 5

HTML Basic

- Structure of an HTML document:

```
<!DOCTYPE html>
<html>
  <head>
    head elements
  </head>
  <body>
    body elements
  </body>
</html>
```

Saved with **.html**
extension

- An HTML document is made up of building blocks called
elements

HTML Elements

- **Elements** are defined by **tags** (markers)
 - Opening tag: `<name>`
 - Closing tag: `</name>`
- The opening tag and its closing tag together specify a **container** for the **content** they enclose
- Not all tags have content: if a tag has **no content**, its form is `<name/>`
- The container and its content together are called an **element**
- Elements can be **nested** (element contains elements)

```
<p>This is a paragraph</p>  
  
```

HTML Attributes

- An HTML element can have **attributes**
- Attributes provide **additional information** about an element
- Attributes are always specified in the **start tag**
- Attributes usually come in name/value pairs like:

name="value"

- Example:

```

```

Convention: use lowercase for attributes

HTML Document Head

- Contains **metadata** of the document: document title, character set, styles, links, scripts, and other meta information
- Defined by the `<head>` tag
- Placed between the `<html>` and `<body>` tags
- Typical tags used to describe the metadata: `<title>`, `<style>`, `<meta>`, `<script>`, and `<base>`

https://www.w3schools.com/html/html_head.asp

HTML Document Body

- Define the content of the document
- Basic content element
 - Headings: `<h1>`, `<h2>`, ..., `<h6>`
 - Paragraphs: `<p>`
 - Quotations: `<blockquote>`
 - Links: `<a href="...">`
 - Images: ``
 - Tables: `<table>`, `<tr>`, `<th>`, `<td>`
 - Lists: `<ol>`, `<ul>`, `<li>`, `<dl>`, `<dt>`, `<dd>`
 - Blocks: `<div>`, `<span>`

HTML Formatting Elements

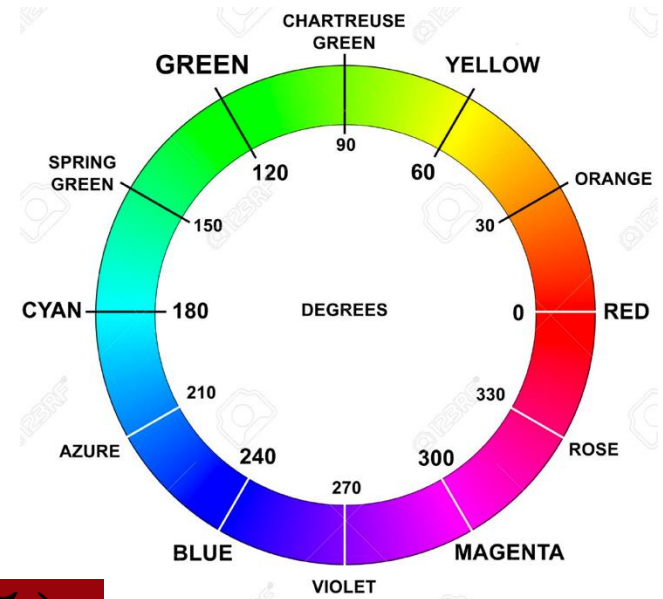
- `<b>`: **Bold text**
- `<strong>`: **Important text**
- `<i>`: *Italic text*
- `<em>`: *Emphasized text*
- `<small>`: Small text
- `<del>`: ~~Deleted text~~
- `<ins>`: Inserted text
- `<sub>`: Subscript text
- `<sup>`: Superscript text
- `<font color="..." face="..." size="...">`: color and font setting
- `<pre>`: pre-formatted text
- `<tt>`: typewriter-style text
- `<code>`: code style (monospace font)
- `<address>`: address section (usually italic)

HTML Styles

- Used to specify the style of an HTML element
- Can be done with the **style attribute**
- Syntax: `<tagname style="property:value;">`
 - `background-color`: specifies background color
 - `color`: specifies text colors
 - `font-family`: specifies text fonts
 - `font-size`: specifies text sizes
 - `text-align`: specifies text alignment

HTML Colors

- Predefine color names: RED, GREEN, BLUE, BLACK, WHITE, etc.
(HTML supports 140 color names)
- HEX: #ffff00 (yellow)
- RGB: rgb(255, 255, 0)
- RGBA (RGB with opacity):
rgba(255, 0, 0, 0.3) 0.5)
- HSL (hue, saturation, light):
hsl(0, 100%, 50%) 80%) 30%)
- HSLA (hue, saturation, light, alpha)



HTML Forms

- Used to **collect data** from the web users
- Form data is **posted to web server** for processing (by an application server such as CGI, ASP, PHP, etc.)
- Tag: `<form action="server-side-script" method="POST/GET">`

Personal Details

Name:

Password:

E-mail id:

Gender: ☐ Male ☐ Female

Contact#:

Educational Qualification

Degree:

Engineering:

Hobbies: ☐ Playing chess ☐ Reading Books

Address

Attach Resume: No file chosen

HTML Forms Elements

- `<input type="...">`: used to **accept data from users**
 - button: a push **button** with no default behavior
 - checkbox: **multiple values** selection
 - file: **file** selection (to upload image to server)
 - image: graphical submit **button**
 - password: **password** input (value is obscured)
 - radio: **single** value selection
 - reset: **resets** the content of the form
 - submit: **submit** the form
 - text: single line **textfield**

HTML5: color, date, email, number, range, search, tel, time, url, week

HTML Forms Elements

- `<button type="..." onclick="...">`: clickable button with predefined standard functions
 - type: submit, reset, button
- `<select>`: provide a menu of options
- `<option>`: define items in a select element
- `<textarea rows="...", cols="...">`: multi-line plaintext editor

Questions

Appendix

References

- <http://www.w3.org>(official)
(http: // www. w3. org/ community/ webed/ wiki/ Main_ Page)
- <http://w3school.com/html/>(recommended)
- <http://htmldog.com/guides/html/>
- <http://www.echoecho.com/html.htm>

Quiz

1. **What does HTML stand for?**
 - a. Hyperlinks and Text Markup Language
 - b. Home Tool Markup Language
 - c. Hyper Text Markup Language

2. **Who is making the Web standards?**
 - a. The World Wide Web Consortium
 - b. Mozilla
 - c. Microsoft
 - d. Google

Quiz

3. Choose the correct HTML tag for the largest heading

- a. `<h1>`
- b. `<heading>`
- c. `<head>`
- d. `<h6>`

4. Choose the correct HTML tag to make a text bold

- a. `<bold>`
- b. `<b>`
- c. `<i>`
- d. `<u>`

Quiz

5. What is the correct HTML for creating a hyperlink?

- a. `<a>http://www.w3schools.com<a>`
- b. `<a url="http://www.w3schools.com">W3Schools.com</a>`
- c. `<a name="http://www.w3schools.com">W3Schools.com</a>`
- d. `<a href="http://www.w3schools.com">W3Schools</a>`

6. How can you create an e-mail link?

- a. `<mail>xxx@yyy</mail>`
- b. `<a href="mailto:xxx@yyy">`
- c. `<mail href="xxx@yyy">`
- d. `<a href="xxx@yyy">`

Quiz

7. How can you open a link in a new browser window?

- a. `<a href="url" target="_blank">`
- b. `<a href="url" target="new">`
- c. `<a href="url" new>`

8. Which of these tags are all table tags?

- a. `<thead>`, `<body>`, `<tr>`
- b. `<table>`, `<tr>`, `<td>`
- c. `<table>`, `<thead>`, `<tfoot>`
- d. `<table>`, `<tr>`, `<td>`

Quiz

9. **What is the correct HTML for inserting an image?**

- a. ``
- b. `<img href="image.gif" alt="MyImage"/>`
- c. `<image src="image.gif" alt="MyImage"/>`
- d. `<img alt="MyImage">image.gif</img>`

10. **Why should you add alternative text to your images?**

- a. So the user can save the image using the text as a name
- b. So the users can get an idea of what the image is before it loads
- c. In case the user wishes to load a different picture
- d. So the users can get an idea of what the image is in case the browser fails to load the image

Quiz

11. To separate single list items use?
- a. `<ul>`
 - b. `<li>`
 - c. `<ol>`
12. When making bulleted lists you have what options?
- a. triangle, square, circle
 - b. square, disc, polygon
 - c. disc, circle, square

Quiz

13. **What are the fields that allows the visitor to enter information called?**
- a. Meta tags
 - b. Form fields
 - c. Meta fields
14. **The value setting of a text field defines what?**
- a. The length of the field
 - b. If the value entered is a proper value
 - c. What will appear in the field as the default value

Quiz

15. **Which field can hold information that does not show?**
- a. Text field
 - b. Hidden field
 - c. Frame field
16. **Password fields are similar to what?**
- a. Hidden fields except text show as “*”
 - b. Text fields except text show as “*”
 - c. Text area except text show as “*”

HTML5

- The latest version of the HTML markup language (5.0, 5.1, 5.2)
- The living standard
- The next generation features for modern web development
- Theoretically, the HTML5 specification will be completed in 2022.

$$\text{HTML5} \sim = \text{HTML} + \text{CSS} + \text{JS}$$

HTML5 Characteristics

- Simpler and cleaner
minimize syntax for tags and attributes
- New structural and semantic elements
richer, more meaningful and more flexible
- New form elements and attributes
13 new input types and many new form elements
- New other elements
multimedia content (video, audio, canvas, menu, etc.)
- New functions
web storage, real-time communication, drag and drop, geolocation, etc.